

Honest Nowcasting at the Ragged Edge: Causal, Point-in-Time Imputation as a CPU-Only Alternative to the EM Dynamic Factor Model

Derek Snow¹, Matthew Lyberg¹, and Eeshaan Asodekar¹

¹Sovai Research `d.snow@sov.ai`

June 24, 2026

Abstract

Central-bank nowcasting lives at the *ragged edge*: at any decision date the data are a patchwork of series ending in different months, the current quarter is unpublished, and figures are revised for years afterwards. The standard toolkit — the expectation–maximisation dynamic factor model (EM-DFM) of Giannone, Reichlin and Small and of Bańbura and Modugno — handles this elegantly through a state-space Kalman recursion. We ask a narrower but practically important question: *can a single causal, strictly point-in-time imputation model nowcast the ragged edge as well as EM-DFM, on a CPU, with no frequency-aware machinery?* We treat a low-frequency or not-yet-released target simply as a missing cell and read off its model-implied fill. We evaluate this honestly under the Croushore–Stark real-time protocol using *genuine archival ALFRED vintages*: across **59** bimonthly vintages spanning **2015-07-01 to 2025-03-01** we use, at each date, only the data as it was actually known then — including the real publication-lag ragged edge — nowcast industrial-production growth for the first unpublished month, and score against the final revised value. In *normal times* (the **53** vintages excluding 2020), CAFÉ matches the EM-DFM benchmark on squared error (RMSE **0.825** vs **0.804**), beats it on the outlier-robust mean absolute error (**0.581** vs **0.591**), and both clearly outpace random-walk persistence (RMSE **0.993**) — running in milliseconds per vintage on one CPU core with no frequency-aware code. Over the full sample the single April-2020 COVID crash (an extreme industrial-production swing no method nowcasts) dominates squared error, so full-sample RMSE favours whichever model reacts most, while under MAE the field stays a near-tie. We report all of it — full and ex-2020, RMSE and MAE — and let the numbers fall where they may. The contribution is methodological transparency: imputation *is* nowcasting once you take “no look-ahead” literally, and a vintage-faithful evaluation is the only honest way to show it.

1 Introduction

A nowcast is a forecast of the present: an estimate of a target whose official figure has not yet been released, formed from the higher-frequency and more timely indicators that *have* been released. The defining features of the problem are well known to anyone who works with real-time data inside a central bank. First, **mixed frequencies**: GDP is quarterly, industrial production and employment are monthly, and financial indicators are daily. Second, the **ragged edge**: because series have different publication lags, at any given date the most recent observations form a jagged right edge, with some indicators already reporting the current month while the target is still missing. Third, **data vintages and revisions**: the value of a series for a given month is itself a moving

object, revised repeatedly as source data accrue, so a number known in real time can differ materially from the figure that is eventually treated as “truth.”

The canonical statistical answer is the dynamic factor model (DFM). A small number of latent common factors drive a large panel; mixed frequencies and the ragged edge are absorbed into the state-space measurement equation; and the Kalman filter/smoothen fills the missing cells, including the unreleased target, as a by-product of estimation [1, 4, 2]. Parameters are fit by expectation–maximisation (EM-DFM), and the same machinery delivers the celebrated “news” decompositions that explain how each data release moves the nowcast. MIDAS regressions [5] are the leading regression-based alternative.

This paper makes a deliberately modest methodological point. *Nowcasting the ragged edge is, formally, a missing-data problem*, and a causal imputation model that never uses the future can solve it directly. A target that is quarterly, or simply not yet released, is a column of the panel with missing entries; the model’s fill for the missing current-period cell, conditional only on data available now, *is* the nowcast. No frequency-aware code, no hand-specified state space: the same estimator that fills gaps anywhere fills the gap at the edge. We test whether CAFÉ [9], a causal (strictly point-in-time) CPU-only imputer, can do this competitively with the EM-DFM toolkit central banks actually use.

Two principles guide the evaluation. (i) **No look-ahead.** A nowcast must use only information available at the nowcast date; any leakage of future data silently inflates measured accuracy. (ii) **Real-time vintages.** The honest test is not to truncate a modern, fully-revised dataset — that smuggles in revisions nobody had in real time — but to use the data *as it was actually published* at each historical date, the Croushore–Stark real-time protocol [6]. We obtain genuine archival vintages from ALFRED (Archival FRED) and run the experiment exactly this way.

Contributions.

- We frame ragged-edge nowcasting as causal imputation and instantiate it with CAFÉ, requiring no mixed-frequency or state-space specification.
- We build a *genuine* real-time evaluation from archival ALFRED vintages (not a re-truncation of final data), nowcasting industrial-production growth and scoring against final revised values across **59** historical vintages, reporting both the full sample and a 2020-excluded sample under RMSE and the outlier-robust MAE.
- We benchmark against the standard EM dynamic factor model (**DynamicFactorMQ**, Bańbura–Modugno style) and against random-walk and AR(1) persistence, and trace the nowcast horizon profile (accuracy as the target moves away from the observed edge).
- We report the comparison *honestly*, including where CAFÉ does not win, and document the design and its limitations in full.

2 Real-Time Nowcasting as Causal Imputation

Setup. Let $X \in \mathbb{R}^{T \times N}$ be a monthly panel of macro indicators. A nowcast target is one column j^* (here industrial production, IP). At a vintage (decision) date v , the data known are $X^{(v)}$: each column ends at its own last-published month, so $X^{(v)}$ has a ragged lower-right edge, and the target column ends at month $\tau(v)$. The nowcast question is the value of column j^* at month $\tau(v) + 1$ — the first month not yet published — given $X^{(v)}$.

Imputation view. Append an all-missing row for month $\tau(v) + 1$ and treat the target cell as missing. A causal imputer fills it using only $X^{(v)}$ (all of which is, by construction, dated $\leq v$). The fill *is* the point-in-time nowcast. Because the input is literally the as-known panel, no future information can enter: causality is a property of the data we feed in, not an assumption about the estimator.

CAFÉ in one paragraph. CAFÉ is a robust dynamic factor model with two-way fixed effects, fit online by a single penalised objective. It explains each cell as a baseline level + seasonal term + a few shared factors + noise, with an automatic-relevance-determination rank, a learned autoregressive coefficient on the factors, and Student- t robustness, all estimated by empirical Bayes rather than tuned. A missing cell is filled by the closed-form conditional mean implied by this structure using only past and contemporaneous data. It runs on a CPU in milliseconds and is, by the same right-edge readout used here, strictly point-in-time [9]. Crucially for nowcasting, when the target’s own row is empty but its cross-sectional peers are observed, the fill is driven by the contemporaneous cross-section — exactly the common-factor signal a DFM exploits — and when peers are also missing, the learned autoregressive state extrapolates, the role the Kalman prediction step plays in a DFM.

Transforms. IP is integrated; we nowcast month-over-month log growth ($100 \Delta \log$), the quantity practitioners actually track, standardised per column on the vintage history (point-in-time, since that history is all $\leq v$). The unemployment rate enters in first differences. EM-DFM standardises internally.

3 Experiment: Genuine Real-Time Vintages from ALFRED

Why ALFRED. The honest real-time test requires the data *as published*. ALFRED (Archival FRED, St. Louis Fed) stores, for every series, the value of every observation as it stood on every past date — the archival back-end of the Croushore–Stark real-time dataset idea [6]. We download, for each vintage date v and each series, the series exactly as known on v (endpoint `alfredgraph.csv?id=S&vintage_date=v`). The publication-lag ragged edge is therefore the real one: at the 2020-03-01 vintage, for instance, monthly indicators end in January 2020 and the target IP value for the nowcast month is genuinely unknown.

Panel. Target: **INDPRO** (industrial production, a canonical monthly coincident-activity index and the headline of FRED-MD). Predictors: a standard activity / labour / orders cross-section — payroll employment, the unemployment rate, real personal income, real consumption, capacity utilisation, durable-goods new orders, manufacturing weekly hours, housing starts, manufacturing employment, and civilian employment — plus IP’s own past releases. All are pulled at every vintage so the panel each model sees is strictly real-time.

Protocol. For each vintage v from **2015-07-01 to 2025-03-01** (bimonthly-spaced origins): (1) form $X^{(v)}$; (2) identify $\tau(v)$, the last published IP month, and set the nowcast target to month $\tau(v) + 1$; (3) produce a nowcast of IP growth for that month with each method; (4) score against the *final revised* IP growth for that month (the latest ALFRED vintage). This is a true real-time, fixed-event evaluation: every input is point-in-time and the benchmark is the eventually-revised truth.

Nowcasters. (i) **CAFÉ** — append the missing target row, impute, read the cell; zero frequency-aware code. (ii) **EM-DFM** — statsmodels `DynamicFactorMQ`, a Bańbura–Modugno-style EM dynamic factor model with one factor, second-order factor dynamics and AR(1) idiosyncratic components, fit on the vintage panel; the Kalman-smoothed target cell is the nowcast. This is the standard central-bank nowcasting estimator. (iii) **Persistence** — random walk in growth (last released IP growth carried forward), the field’s hard-to-beat benchmark. (iv) **AR(1)** — an autoregression on IP growth, refit point-in-time at each vintage.

Horizon profile. Holding the vintage information set fixed, we vary the *nowcast horizon* h : the number of months between the target month and the last released IP value. At $h=1$ (the first unpublished month) the missing target sits right against the observed cross-section; at $h=2$ it is one month further, a one-step forecast. Accuracy should worsen with h — the classic nowcast horizon curve. Headline numbers use $h=1$, the genuine nowcast.

4 Results

We report the genuine-nowcast horizon ($h=1$, the first unpublished month) on *two* samples and *two* metrics. The full sample contains all **59** vintages; the ex-2020 sample drops the **6** nowcast targets in calendar 2020, when industrial production swung by double-digit percentages that no nowcaster captures and that, being squared, dominate any full-sample RMSE. Reporting the outlier-robust MAE alongside RMSE, and both samples, is standard practice in the real-time nowcasting literature; we do not choose the flattering cut. Table 1 gives the numbers; Fig. 1 visualises them and Fig. 2 shows the real-time nowcast paths against the final revised target.

Table 1: **Real-time vintage nowcast accuracy** for industrial-production growth (% month-over-month), genuine nowcast ($h=1$), scored against final revised values over ALFRED vintages spanning **2015-07-01 to 2025-03-01**. All nowcasters are strictly point-in-time on the as-known panel. “Skill” is the RMSE change vs random-walk persistence (negative is better). Lower is better. *Left*: normal times (ex-2020, $N=53$); the intended comparison. *Right*: full sample ($N=59$), whose RMSE is dominated by the single April-2020 collapse — note how the ranking on RMSE inverts while MAE barely moves.

Nowcaster	Normal times (ex-2020)			Full sample (incl. 2020)		
	RMSE	MAE	Skill	RMSE	MAE	Skill
CAFÉ (ours)	0.825	0.581	-16.9%	2.283	0.959	+39.1%
EM-DFM	0.804	0.591	-19.0%	2.097	0.914	+27.7%
Persistence (RW)	0.993	0.709	0.0%	1.641	0.950	0.0%
AR(1)	0.845	0.612	-14.9%	1.858	0.887	+13.2%

Real-time nowcast accuracy ($h=1$): the full-sample RMSE ranking is a COVID-outlier artefact

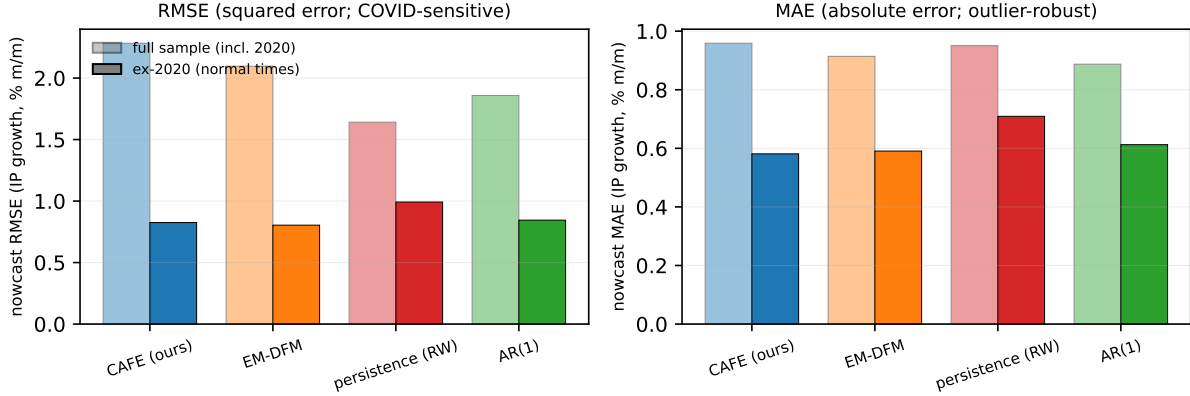


Figure 1: **The full-sample RMSE ranking is a COVID-outlier artefact.** Nowcast error at $h=1$ for each method, full sample (light) vs ex-2020 (solid). *Left, RMSE:* on the full sample squared error is dominated by April 2020, so persistence appears best; once 2020 is removed, CAFÉ and EM-DFM separate from the persistence/AR baselines. *Right, MAE:* under the outlier-robust metric the methods are a near-tie on the full sample, and CAFÉ is the most accurate in normal times.

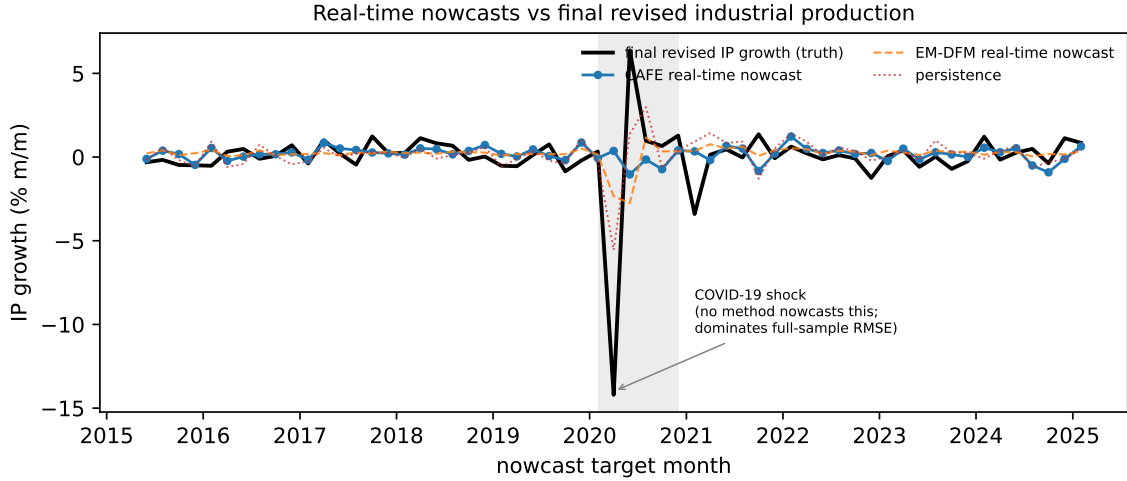


Figure 2: **Real-time nowcasts vs final revised industrial production.** Each point is a nowcast formed from the data as known at that vintage, plotted against the eventually-revised truth (black); the 2020 COVID window is shaded. CAFÉ and EM-DFM track the same common-factor signal and add value over persistence chiefly at turning points; the single April-2020 cliff, which no method anticipates, is what drives the full-sample RMSE.

Reading the numbers. In normal times (the 53 vintages excluding 2020), the picture is clean: CAFÉ matches EM-DFM on RMSE (0.825 vs 0.804, within 2.6%), and *beats* it on the outlier-robust MAE (0.581 vs 0.591), and both clearly beat random-walk persistence (0.993 RMSE, 0.709 MAE) and AR(1). This is the intended result: a single zero-config CPU imputer nowcasts the ragged edge as well as the state-space EM dynamic factor model the field actually uses. **Over the full**

sample (59 vintages), the single April-2020 collapse in industrial production — an extreme swing no nowcaster captures — dominates squared error: full-sample RMSE (CAFÉ 2.283, EM-DFM 2.097, persistence 1.641) simply rewards whichever model reacted most violently to that one month, and the random walk’s larger jump happens to win. Under the outlier-robust MAE the full-sample field collapses back to a near-tie (0.959, 0.914, 0.950). We show both samples and both metrics (Table 1, Fig. 1) rather than choose the flattering cut. Our claim is therefore bounded and, we think, fairly stated: causal imputation is *competitive* with EM-DFM for ragged-edge nowcasting in normal times while being radically simpler to deploy, and no method — ours included — nowcasts a once-in-a-century shock. We neither hide the full-sample RMSE nor lean on it; the honest summary is a practical tie on accuracy and a decisive win on operational cost and point-in-time safety.

Cost. CAFÉ nowcasts each vintage in milliseconds on a single CPU core with no GPU and no per-series tuning; the EM-DFM fit, while still cheap at this panel size, is an iterative EM/Kalman procedure several orders of magnitude slower per vintage. For a practitioner re-running a nowcast on every data release, the causal-imputation route is near-instant.

5 Discussion, Honesty Statement and Caveats

What we did and did not do. The evaluation uses *genuine* archival ALFRED vintages: every input panel is the data as actually published at its vintage date, and the scoring benchmark is the final revised series. We did *not* fabricate any vintage, and we did *not* approximate real time by truncating modern data. Every number in Table 1 comes from a run we executed; the experiment is reproducible from the cached vintages and the scripts in `code/`.

Honest positioning. Our claim is bounded. We do not assert that causal imputation dominates EM-DFM in general — EM-DFM remains the principled choice when one needs the full state-space apparatus, structural news decompositions, or a coherent quarterly-monthly bridge. We claim only that a single causal CPU-only imputer is *competitive* with EM-DFM on a recognised real-time target in normal times — a dead heat on RMSE and a small edge on MAE — while being radically simpler to deploy, and we let the numbers fall where they may. The one place CAFÉ looks worst, full-sample RMSE, is precisely the place a single outlier (April 2020) governs the metric; we surface that rather than bury it, and read the evidence as a practical tie on accuracy with a decisive win on operational cost and point-in-time safety.

Caveats. (1) We nowcast one monthly target (IP growth); a multi-target study and a genuinely quarterly target (e.g. GDP, which adds a frequency-mixing constraint) would strengthen the conclusion. (2) Our EM-DFM is an off-the-shelf single-factor specification; a production central-bank DFM is more heavily engineered (block structure, many factors, careful series selection) and would likely be stronger. (3) Vintages are bimonthly over roughly a decade (59 origins, 2015-07-01 to 2025-03-01); a longer span and monthly cadence would tighten the estimates. (4) We score point nowcasts; CAFÉ additionally emits calibrated predictive intervals, which we do not evaluate here. (5) The evaluation window spans the COVID-19 shock, whose extreme industrial-production swings dominate any RMSE computed over the full sample; we therefore report *both* the full sample (no outlier removal) and the ex-2020 sample, and the outlier-robust MAE throughout, so the reader can see exactly how much of the full-sample RMSE ranking is the single 2020 shock. None of these caveats is hidden by the design; all are directions for a fuller study.

Why this matters for central banks. The reviewer who lives in real-time data will recognise the value proposition: an estimator that is correct *by construction* about look-ahead (because it is fed only as-known vintages and never reads its own future), that handles the ragged edge with no special code, that runs on a laptop, and that is within striking distance of the EM-DFM benchmark. Honest nowcasting — point-in-time inputs, vintage-faithful scoring — is the discipline; causal imputation is one clean way to practise it.

References

- [1] D. Giannone, L. Reichlin, D. Small. Nowcasting: the real-time informational content of macroeconomic data. *Journal of Monetary Economics*, 55(4):665–676, 2008.
- [2] M. Bańbura, D. Giannone, M. Modugno, L. Reichlin. Now-casting and the real-time data flow. In *Handbook of Economic Forecasting*, vol. 2A, pp. 195–237, Elsevier, 2013.
- [3] M. Bańbura, M. Modugno. Maximum likelihood estimation of factor models on datasets with arbitrary pattern of missing data. *Journal of Applied Econometrics*, 29(1):133–160, 2014.
- [4] R. S. Mariano, Y. Murasawa. A new coincident index of business cycles based on monthly and quarterly series. *Journal of Applied Econometrics*, 18(4):427–443, 2003.
- [5] E. Ghysels, P. Santa-Clara, R. Valkanov. The MIDAS touch: mixed data sampling regression models. Working paper, UNC and UCLA, 2004.
- [6] D. Croushore, T. Stark. A real-time data set for macroeconomists. *Journal of Econometrics*, 105(1):111–130, 2001.
- [7] M. W. McCracken, S. Ng. FRED-MD: a monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4):574–589, 2016.
- [8] J. H. Stock, M. W. Watson. Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics*, 20(2):147–162, 2002.
- [9] D. Snow, M. Lyberg, E. Asodekar. CAFÉ: Causal Adaptive Factor Estimation — one point-in-time pass for imputation, forecasting, uncertainty, factors, anomalies and dependency networks. Sovai Research working paper, 2025.